

EXPERIMENT NO. : 3DATE: 13/08/2026

1. AIM

To design and study a lithography mask pattern for shallow trench isolation, illustrating line-space structures within a defined boundary for device isolation.

2. TOOL

Semulator3D

3. THEORY

Shallow Trench Isolation (STI) is a widely used technique in integrated circuit fabrication that provides electrical insulation between closely placed semiconductor devices so that unwanted current leakage does not occur. In this method, shallow trenches are etched into the silicon wafer in the regions where isolation is required. These trenches are then filled with a dielectric material, usually silicon dioxide, which acts as an insulating barrier. After filling, the wafer undergoes a Chemical Mechanical Polishing (CMP) process to remove excess dielectric and achieve a smooth, planar surface so that further fabrication steps can proceed accurately.

STI became the preferred isolation technique for advanced CMOS technologies, particularly for process nodes of 250 nanometers and below, because it allows isolation regions to be made narrower and more precise compared to earlier methods. Before STI became standard, older CMOS nodes and several non-MOS technologies typically relied on LOCOS (Local Oxidation of Silicon) for isolation.

However, LOCOS suffered from issues like the bird's beak effect, which caused lateral oxidation beyond the intended region and reduced packing density. STI overcomes these limitations by providing better control over isolation dimensions and enabling closer placement of devices. It therefore supports improved device scaling, higher packing density and reliable electrical isolation, making it an important isolation technique in modern semiconductor manufacturing.

In lithography, the STI mask pattern defines the regions where the isolation trenches are to be formed. The line-space pattern consists of alternating line and space regions within a defined boundary. Accurate dimensions and spacing of these structures are important for obtaining the desired isolation geometry and maintaining pattern fidelity during subsequent fabrication steps.

4. DESIGN PARAMETERS

Table 1: DESIGN PARAMETERS

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	Silicon/Silicon...		200			WAFER_1				
2		Conformal...	pad oxide deposition	/Oxides/Silicon...		4			WAFER_1	Both	1.0		
3		Conformal...	pad nitride deposition	/Nitrides/SiN		40			WAFER_1	Top	1.0		
4		Sequence	lithography						WAFER_1				
4.1		Planarizing...	spin on resist	/Organics/photo...		120			WAFER_1	Top			
4.2		Expose resist	Expose resist	/Organics/photo...		0	RX	Light	WAFER_1	Top			
5		Sequence	etch fins						WAFER_1				
5.1		Enhanced B...	pad nitride etch	/Nitrides/Org...		40			WAFER_1	Top	0.05		
5.2		Enhanced B...	pad oxide etch	/Nitrides/Org...		4			WAFER_1	Top	0.05		
5.3		Enhanced B...	silicon etch	/Nitrides/Org...		100			WAFER_1	Top	0.05		
6		Remove M...	resist strip	/Organics/					WAFER_1				
7		Conformal...	sti fill deposition	/Oxides/SiO2_H...		200			WAFER_1	Top	0.67		
8		Polish	sti oxide cmp						WAFER_1	Top			

5. RESULT

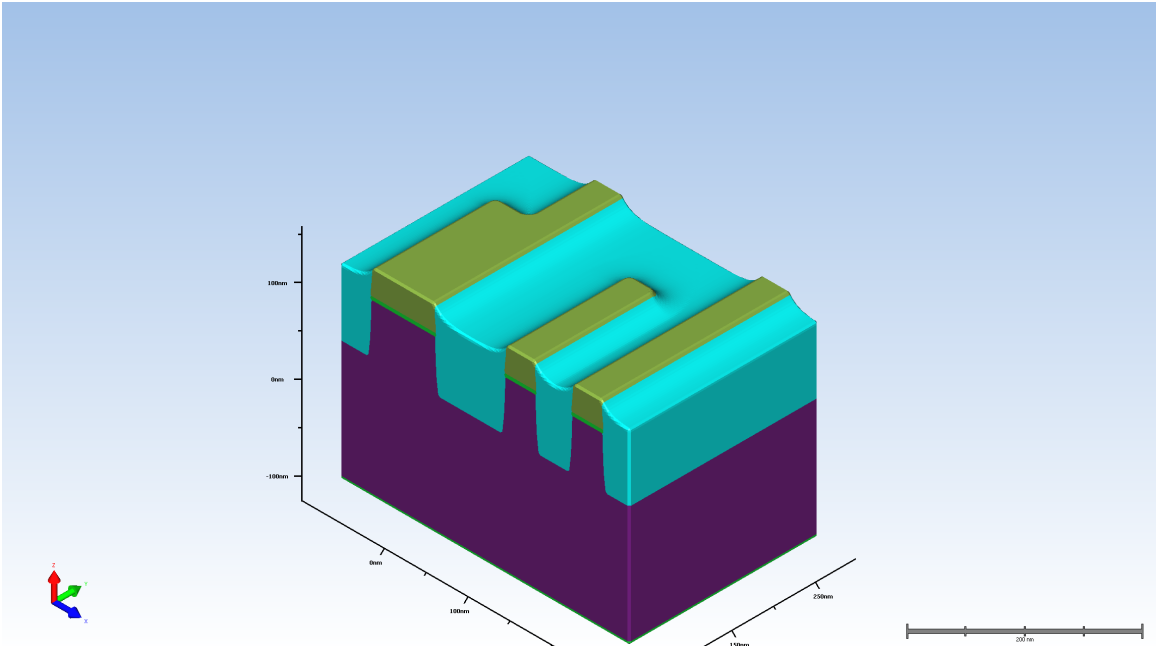


Figure 1: final result

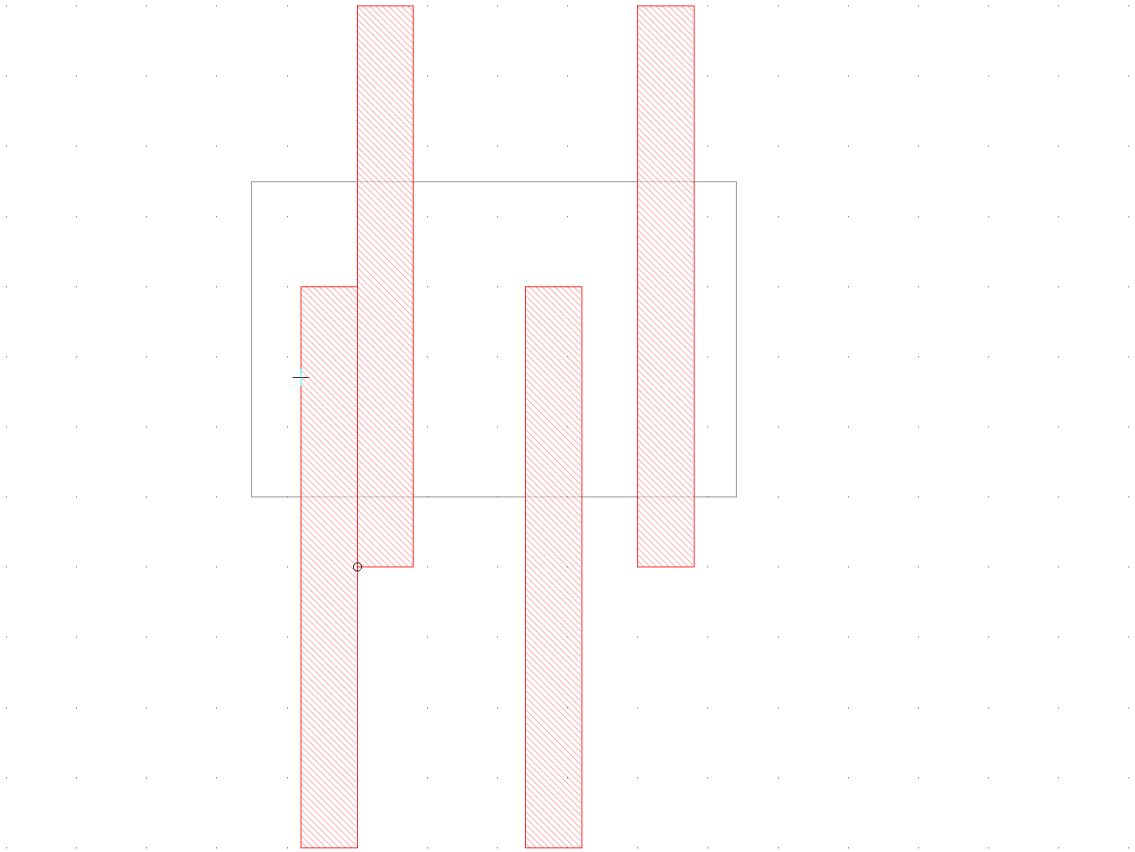


Figure 2: Layout

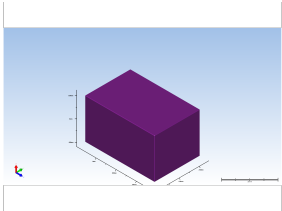


Figure 3

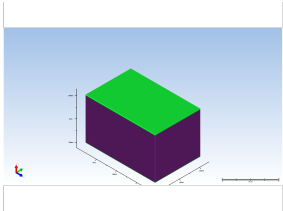


Figure 4

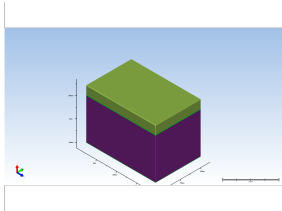


Figure 5

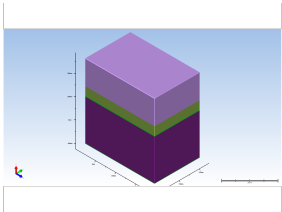


Figure 6

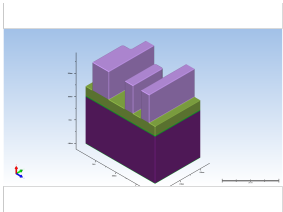


Figure 7

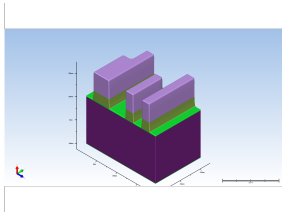


Figure 8

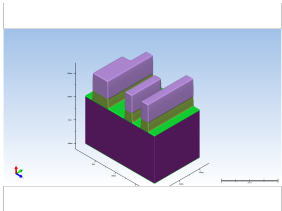


Figure 9

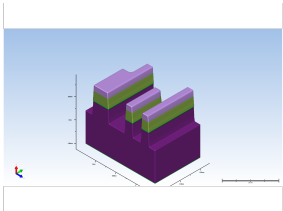


Figure 10

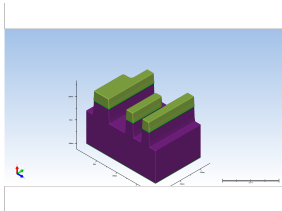


Figure 11

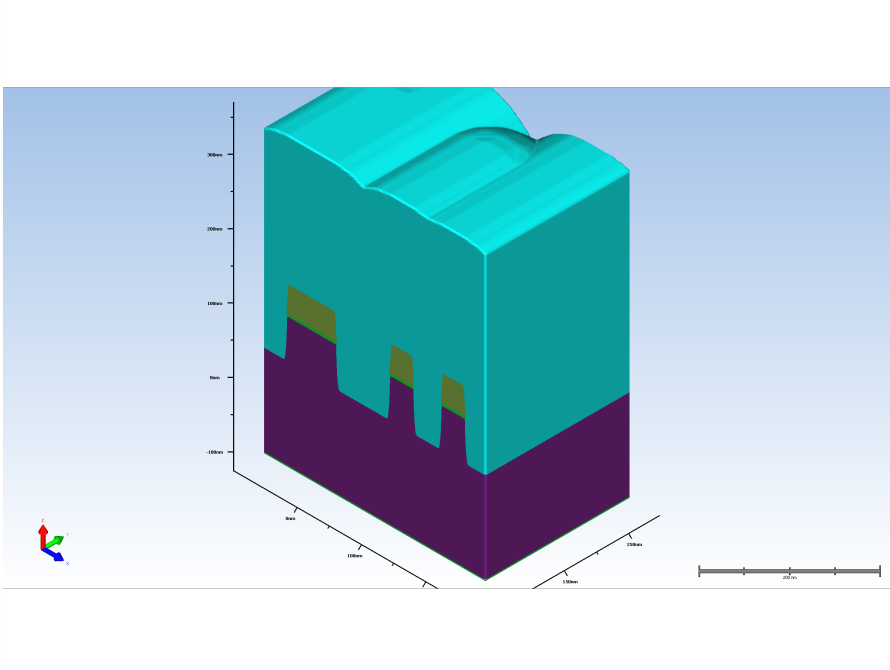


Figure 12

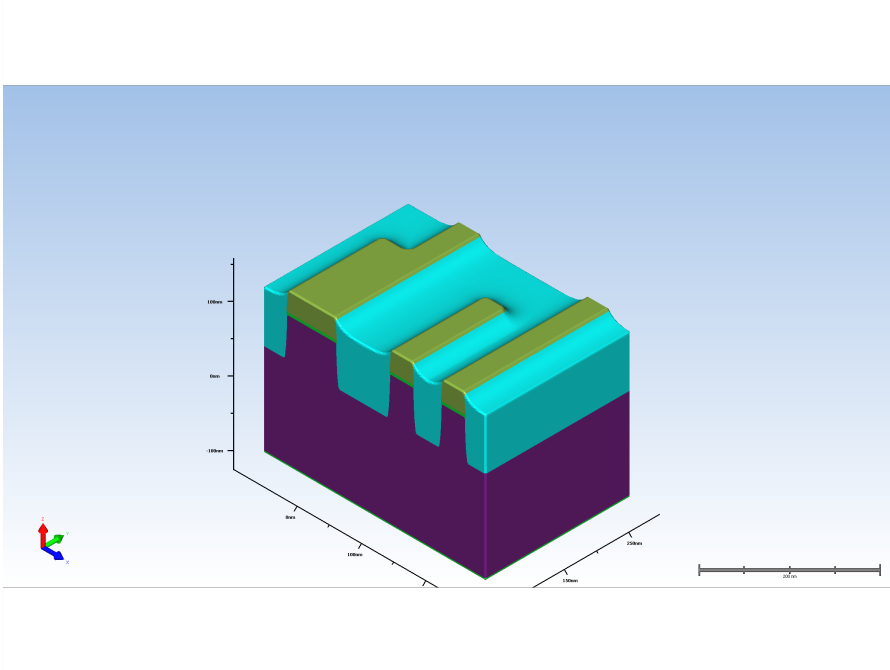


Figure 13: final result

6. OBSERVATION

The lithography mask pattern was designed with alternating line and space regions within the defined boundary, representing the required shallow trench isolation pattern.

7. CONCLUSION

The shallow trench isolation lithography mask pattern was successfully designed and studied using Simulator3D. The line-space structures were created within the specified boundary to represent the regions required for device isolation. The experiment provided an understanding of how lithography patterns are used to define isolation regions during semiconductor fabrication.